FNCE30007 DERIVATIVE SECURITIES
Tutorial 5 Questions

1. Consider the one period binomial model for a stock price. Using the following results presented in lectures:

$$\Delta u S_0 - f_u = \pi$$

$$\Delta S_0 - f = e^{-rT}\pi$$

Where f_u, f_d and f are the option values in the up state, down state and t=0 respectively, and $\Delta = \frac{f_u - f_d}{S(u-d)}$.
Show that: $f = e^{-r\Delta t}[p * f_u + (1-p)f_d]$
where $p = \frac{e^{r\Delta t} - d}{u-d}$

2. A stock price is currently \$50. In each of the next two three-month periods, the stock price has an equal probability of increasing by 6% or decreasing by 5%. The risk-free interest rate is 5% per annum with continuous compounding. What is the value of a six-month European call option with a strike price of \$51? Check your answer using the simulator.

3. For the situation considered in Question 2, what is the value of a six-month European put option with a strike price of \$51? Verify that the European call and European put prices satisfy put–call parity. Check your answer using the simulator.

4. How do the answers to question 2 and 3 change if the probability of an up-move increases to 55% but all other details remain the same?

5. A stock price is currently \$50. It is known that at the end of two months it will be either \$53 or \$48. The risk-free interest rate is 10% per annum with continuous compounding. What is the value of a two-month European call option with a strike price of \$49? Use no-arbitrage arguments.